

Test Plan and Execution Report

Student Club Event Ticketing and Budget Portal

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This report covers the student club portal built in this repository. It is separate from the lecturer-supplied game exercise, whose repository and four test cases were not supplied.

1 Test setup

Tests ran locally on Windows using Python 3.12, Flask, SQLite and pytest. Each automated test uses a fresh temporary database. The full package list and machine details are stored in results/python-packages.txt and results/performance.json. Demo data is synthetic.

2 Scope and method

The tests check the business services and HTTP routes. They cover both normal input and invalid actions, including concurrency for budget reservations, the last seat and duplicate entry. The browser walkthrough uses the actual running portal. No screenshot is drawn to imitate an application.

3 Execution summary

Run	Observed result	Evidence
Initial regression run	19 passed, 1 failed	results/initial-tests.txt and initial-tests.xml
After the ticket fix	20 passed	results/final-tests.txt and final-tests.xml
Patch	Reject malformed signatures before comparison	results/ticket-validation-fix.patch

4 Defect DEF01 Malformed ticket signature

Input: a ticket string whose signature contains non-ASCII characters. Expected: the system rejects it as invalid. Actual before the fix: Python raised TypeError during hmac.compare_digest. This could produce an error page instead of the expected rejection.

Fix: require a 64-character lowercase hexadecimal signature before comparing it with the expected HMAC. TC17 now passes. The full test suite was rerun to check that valid tickets and the other rejection cases still worked.

5 Test cases

ID	Check	Final result
TC01	Missing fields and invalid money	Passed
TC02	Draft and budget item storage	Passed
TC03	Approval order and recorded sign-offs	Passed
TC04	Rejection reason and allocation release	Passed
TC05	Over-budget request and draft revision	Passed
TC06	Registration before approval	Passed
TC07	Ticket uniqueness and repeat registration	Passed
TC08	Event capacity	Passed
TC09	Single-use entry	Passed
TC10	Tampering, wrong event and expiry	Passed
TC11	Roles and ownership	Passed
TC12	Concurrent scans accept once	Passed
TC13	Concurrent requests do not overspend	Passed
TC14	Status API reflects committed changes	Passed
TC15	Login, CSRF, logout and authentication	Passed
TC16	Full HTTP flow and QR response	Passed
TC17	Malformed non-ASCII signature	Passed
TC18	Concurrent registration for the last seat	Passed
TC19	Draft edits and locking after submission	Passed
TC20	HTML escaping and SQL login input	Passed

6 Performance measurements

Run time (UTC): 2026-10-04T18:19:07.447073+00:00. Loopback HTTP against waitress; 500 distinct pre-authenticated users released together. Server HTML response time, not browser paint or WAN latency. QR check-in POST includes CSRF, authorization and database update.

Workload	Successful	Median ms	P95 ms	Max ms	Target met
500 concurrent page requests	500/500	1287.16	1946.67	2167.95	No
50 sequential check-ins	50/50	27.44	29.61	33.97	Yes
50 check-ins using 20 workers	50/50	171.32	411.5	715.13	No

Targets: under 2,000 ms for each page load and under 100 ms for each check-in. These measurements include local HTTP processing but not QR hardware acquisition, browser rendering or a real network. The 500-user run uses distinct pre-authenticated users; it measures page reads, not 500 simultaneous registrations.

The final run uses Waitress with 16 worker threads and direct loopback requests. Earlier runs used the development server or the default Windows proxy lookup; their raw results are retained as diagnostic baselines and are not directly comparable. The direct connection removes proxy discovery from the local measurement.

All 500 final page requests succeeded, but some exceeded two seconds. Sequential check-ins met 100 ms, while the concurrent check-in run did not. NFR-001 and NFR-002 are therefore not fully met. Further work should investigate database write contention and measure the application on the intended infrastructure; the present result is a local prototype benchmark.

7 Manual checks and open items

Real browser screenshots record proposal creation, approval history, ticket display and check-in. A narrow viewport review checks layout only; it does not replace testing on a physical phone. The saved screenshots and steps are in 3-Project-Evidence.

Open items: real scanner testing, physical mobile-device testing, browser page-load measurements on target infrastructure, and the separately assigned game repository. The GitHub Copilot and Jira evidence also remains dependent on those tools being available.